Fine-Tuning LLMs and RLHF

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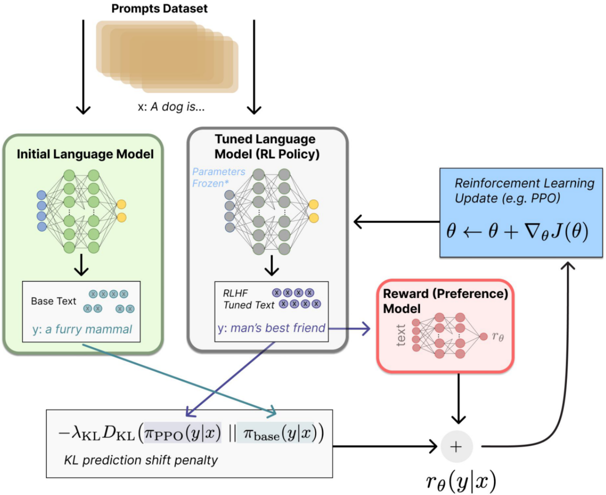

Prompts Dataset
x: A dog is...
Initial Language Model
Base Text
y: a furry mammal
Tuned Language Model (RL Policy)
Parameters Frozen*
RLHF Tuned Text
y: man's best friend
Reinforcement Learning Update (e.g. PPO)
$\theta \leftarrow \theta + \nabla_\theta J(\theta)$
Reward (Preference) Model
text
r_θ
$-\lambda_{\text{KL}} D_{\text{KL}}(\pi_{\text{PPO}}(y|x) \,||\, \pi_{\text{base}}(y|x))$
KL prediction shift penalty
+
$r_\theta(y|x)$

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Table of Contents (Detailed Outline)

- Pre-trained LLMs are powerful but not aligned to specific use-cases or human preferences.

Motivating Examples:

- GPT-3 $\rightarrow$ InstructGPT
- LLaMA $\rightarrow$ Alpaca, Vicuna
- ChatGPT & Claude $\rightarrow$ RLHF-tuned

Motivation

- Pre-trained LLMs are powerful but not aligned to specific use-cases or human preferences.
- **We need:**
  - Adaptability (domain tuning)
  - Alignment (user intention)
  - Efficiency (cost-effective methods)

Motivating Examples:

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Motivation

- Pre-trained LLMs are powerful but not aligned to specific use-cases or human preferences.
- **We need:**
  - Adaptability (domain tuning)
  - Alignment (user intention)
  - Efficiency (cost-effective methods)
- Fine-tuning and RLHF bridge the gap between general intelligence and practical usability.

Motivating Examples:

- ▷ GPT-3 → InstructGPT
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- ▷ ChatGPT & Claude → RLHF-tuned

Learning Outcomes

After this session, you will be able to:

- Explain and compare different fine-tuning methods for LLMs
- Understand the pipeline of Supervised Fine-Tuning and RLHF
- Describe PPO (Proximal Policy Optimization) and DPO (Direct Preference Optimization)
- Apply Low-Rank Adaptation (LoRA) and Quantized LoRA for efficient tuning
- Analyze trade-offs, limitations, and future directions of LLM adaptation

Fine-Tuning Methods for LLMs

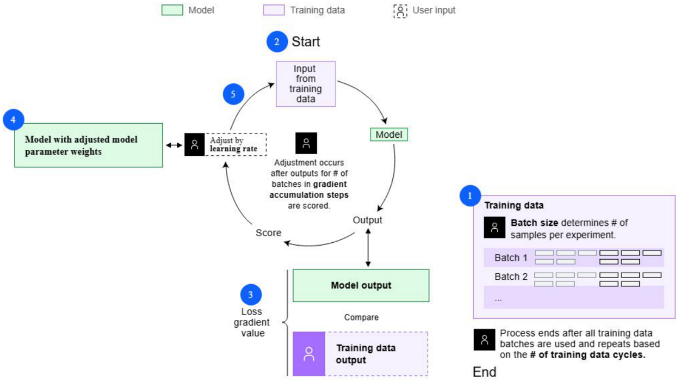
Categories of Fine-Tuning

Method	Description	Params	Efficiency
Full FT	Retrain all weights	All	Expensive
Adapter	Add small bottlenecks (Houlsby)	Few	Efficient
Prefix	Tune soft prompts	Few tokens	Efficient
LoRA/QLoRA	Low-rank weight deltas	Very few	Very Efficient
Instr. Tuning	Tune on instr. datasets	All/Part	-

Full Fine-Tuning (FT)

- **Definition:** Fine-tune all parameters of the pre-trained model on downstream data.
- **Historical Use:** Common in early GPT-2 and BERT applications.
- **Pros:**
  - Maximum flexibility and performance
- **Cons:**
  - Expensive (requires large compute resources)
  - Prone to catastrophic forgetting
  - Not efficient for large models

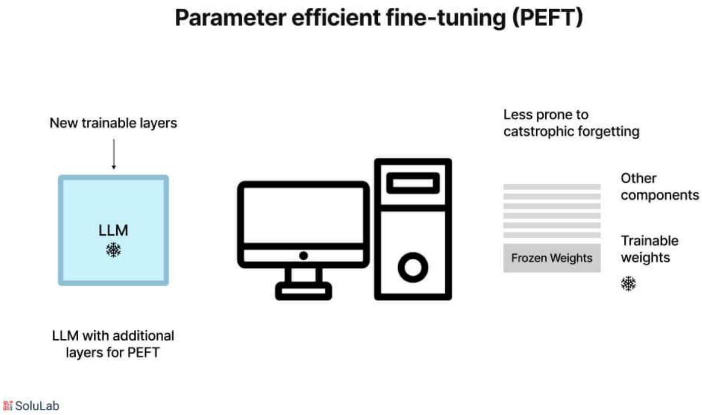
Full Fine-Tuning Workflow



Parameter-Efficient Fine-Tuning (PEFT)

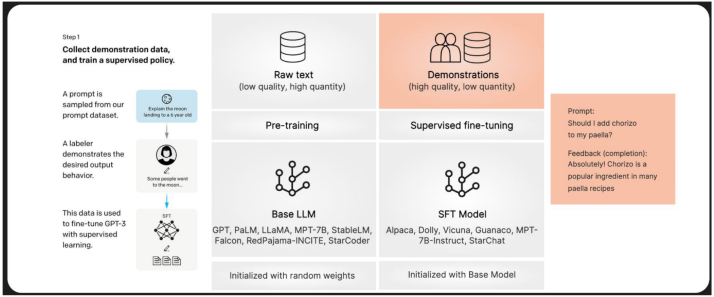
- **Key Idea:** Keep the base model frozen, tune only a small subset of parameters.
- **Types:**
  - Adapter Modules (Houlsby et al., 2019)
  - Prompt Tuning / Prefix Tuning
  - LoRA / QLoRA
- **Benefits:**
  - Enables multi-tasking and personalization
  - Suitable for low-resource adaptation
  - Reduces compute and memory requirements

Parameter-Efficient Fine-Tuning (PEFT)

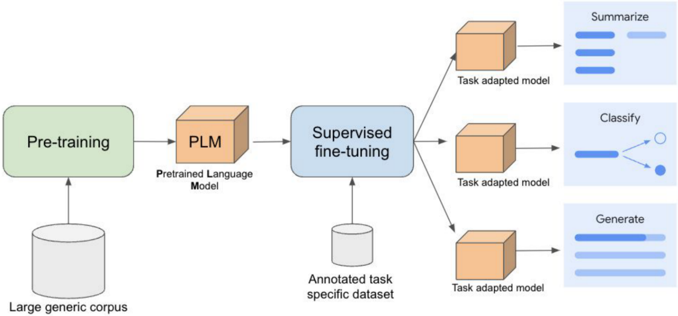


Supervised Fine-Tuning (SFT)

What is Supervised Fine-Tuning (SFT)?



Supervised Fine-Tuning For Gemini LLM



Supervised Fine Tuning for Gemini LLM

Datasets for Supervised Fine-Tuning (SFT)

- **Instructional datasets:**
  - **OpenAI:** InstructGPT (Anthropic Helpfulness data)
  - **Stanford Alpaca:** 52k GPT-3 generated instructions
  - **Others:** Dolly, ShareGPT, OASST, UltraChat
- **Note:** Quality of supervision greatly affects model behavior.

Limitations of Supervised Fine-Tuning (SFT)

- Cannot capture nuanced human preferences or values.
- May reinforce existing biases or hallucinations present in the data.
- Risk of overfitting, especially on synthetic or noisy datasets.
- Often leads to safe but bland and generic responses.

RLHF — Reinforcement Learning with Human Feedback

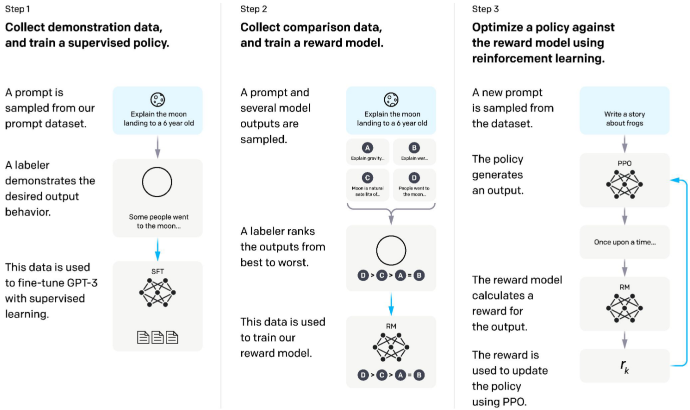
Why RLHF?

- SFT doesn't teach the model what humans prefer.
- RLHF introduces reward learning and policy optimization.
- Inspired by InstructGPT (Ouyang et al., 2022).

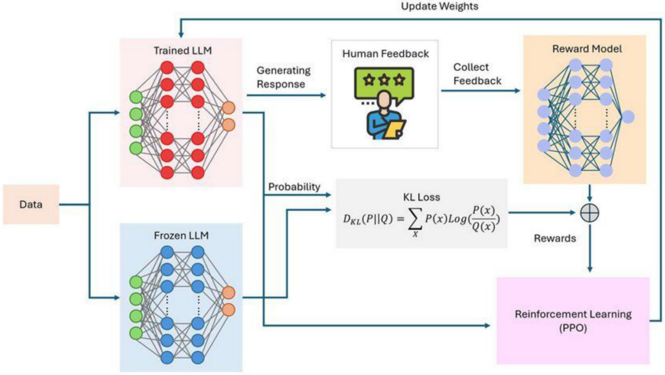
RLHF Pipeline Overview

1. **Supervised Fine-Tuning (SFT):** Train the base model on instruction-response pairs.
2. **Reward Model (RM) Training:** Learn a reward function from human preference rankings.
3. **RL Optimization (e.g., PPO):** Optimize the language model to maximize the learned reward.

RLHF Pipeline



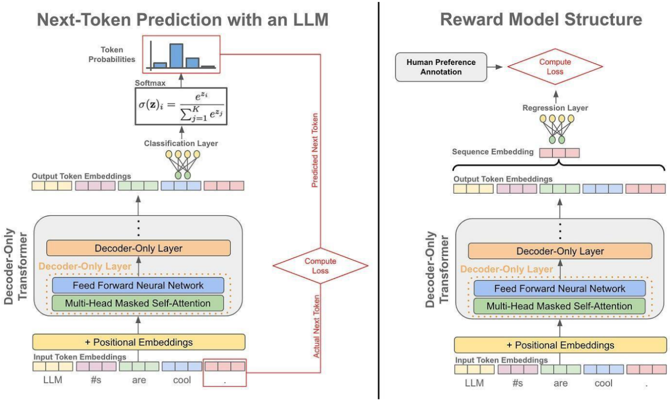
RLHF Pipeline



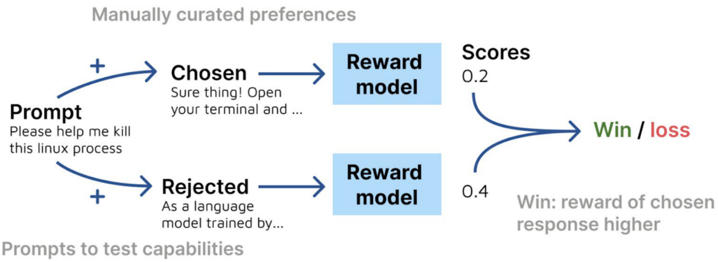
Reward Model (RM)

- **Given:** Two responses y_A and y_B to the same prompt.
- **Human Feedback:** Annotator selects which response is preferred.
- **Learning:** The reward model r_ϕ is trained to assign higher scores to preferred responses.
- **Pairwise Loss:**
$$L_{\mathrm{RM}} = -\log \sigma\Big(r_\phi(y_A) - r_\phi(y_B)\Big)$$
  where σ is the sigmoid function.
- The RM acts as a proxy for human judgment in downstream RL optimization.

Model Fine-tuning for RLHF



Model Fine-tuning for RLHF



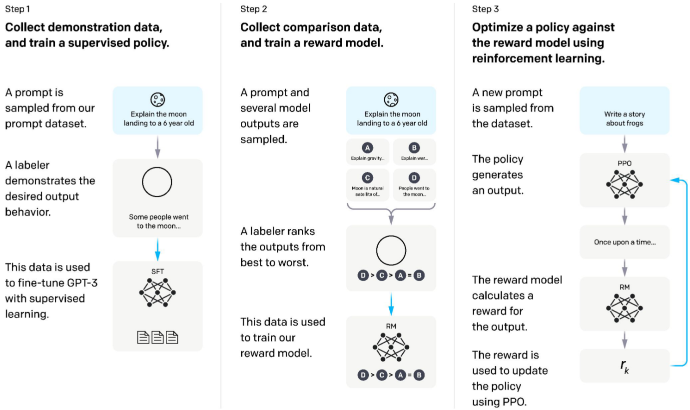
- **Objective:** Maximize the reward given by the reward model (RM).
- **Algorithm:** Proximal Policy Optimization (PPO) is used to update the language model.
- **PPO Loss:**

$$L_{\mathrm{PPO}} = \mathbb{E}\left[\min\left(r_t(\theta)\hat{A}_t, \quad \mathrm{clip}\left(r_t(\theta), 1-\epsilon, 1+\epsilon\right)\hat{A}_t\right)\right]$$

  where $r_t(\theta) = \frac{\pi_\theta(a_t|s_t)}{\pi_{\mathrm{old}}(a_t|s_t)}$ is the probability ratio, and $\hat{A}_t$ is the advantage estimate.
- **KL Penalty:** A KL-divergence penalty is added to keep the policy close to the base (SFT) model:

$$L_{\mathrm{KL}} = \beta\,\mathrm{KL}\left(\pi_\theta \parallel \pi_{\mathrm{SFT}}\right)$$

RLHF Training: Proximal Policy Optimization (PPO)



PPO — Pros and Cons

- ✓ Stable optimization
- ✓ Encourages exploration
- × Expensive (needs many rollouts)
- × Sensitive to reward model errors

- **New method:** Bypasses RL and reward model training entirely.
- **Key Idea:** Optimize the policy directly from human preference data.
- **Reference:** Rafailov et al., 2023.

DPO Loss:

$$L_{\mathrm{DPO}} = \log \frac{\exp\left(\beta \log \frac{\pi(y^+)}{\pi_0(y^+)}\right)}{\exp\left(\beta \log \frac{\pi(y^+)}{\pi_0(y^+)}\right) + \exp\left(\beta \log \frac{\pi(y^-)}{\pi_0(y^-)}\right)}$$

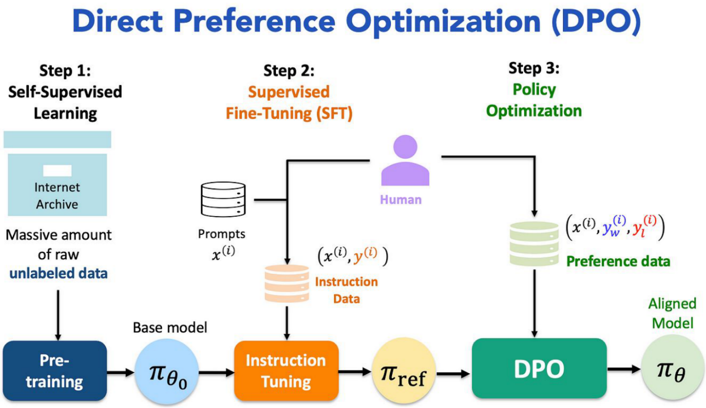
where y^+ is the preferred response, y^- is the dispreferred response, π is the current policy, π_0 is the reference (SFT) policy, and β is a temperature parameter.

Direct Preference Optimization (DPO) (cont.)

Benefits:

- Simpler and more stable than PPO
- No need for reward model training or RL rollouts

Direct Preference Optimization (DPO)



LoRA and Quantized LoRA

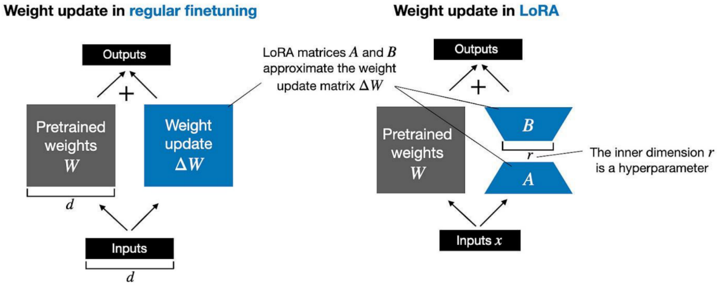
What is LoRA?

- **Key Idea:** Update low-rank matrices instead of full weights.
- For a weight matrix $W \in \mathbb{R}^{d \times k}$:

$$W' = W + \Delta W, \quad \Delta W = AB^T$$

  where $A \in \mathbb{R}^{d \times r}$, $B \in \mathbb{R}^{k \times r}$.
- Only train $A, B \rightarrow$ drastically reduces parameters.

What is LoRA?


Weight update in regular finetuning
Outputs
+
Pretrained weights W
Weight update ΔW
d
Inputs
d
LoRA matrices A and B approximate the weight update matrix ΔW
Weight update in LoRA
Outputs
+
Pretrained weights W
B
r
A
The inner dimension r is a hyperparameter
Inputs x

Benefits of LoRA

- **Very lightweight:** Only 0.1%–1% of parameters are trainable.
- **Hardware friendly:** Enables fine-tuning on consumer GPUs and even laptops.
- **Modular:** Supports plug-and-play adapters for different tasks or users.
- **Personalization:** Allows efficient user- or domain-specific adaptation.

QLoRA — Quantized LoRA

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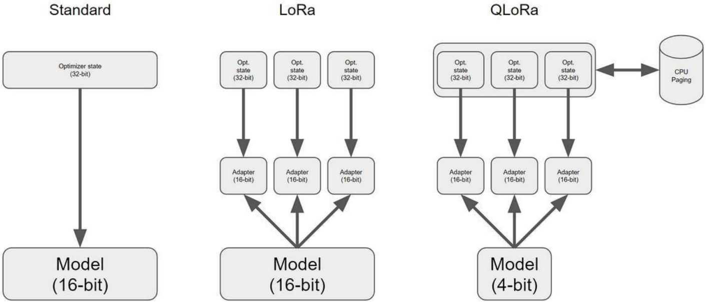
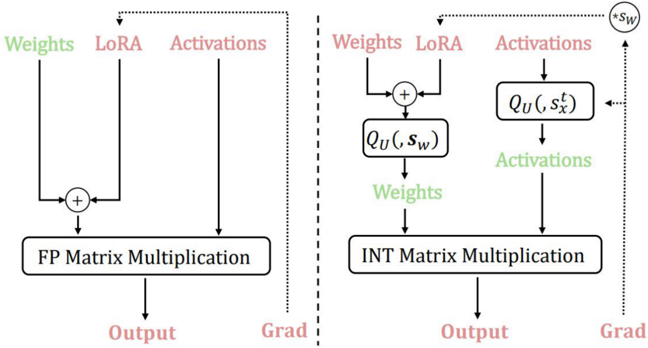
- **Key Idea:** Fine-tune large language models in 4-bit precision using LoRA adapters.
- **Scalability:** Enables fine-tuning of 65B parameter models on a single 48GB GPU.
- **Efficiency:** Highly memory-efficient with no significant performance loss (Dettmers et al., 2023).
- **Techniques:**
  - Double quantization
  - Paged optimizers
  - NF4 (NormalFloat 4-bit) quantization format

QLoRA — Quantized LoRA



QLoRA — Quantized LoRA



(a) QLoRA layer (b) QALoRA layer

Applications of LoRA / QLoRA

- **Instruction tuning for resource-constrained settings:**
  - Enables fine-tuning large models on modest hardware (e.g., consumer GPUs, laptops).
  - Used for domain adaptation, personalization, and rapid prototyping.
- **Popular fine-tuned models:**
  - **Alpaca**, **Guanaco**, **Vicuna**, **Mistral**: All leverage LoRA/QLoRA for efficient instruction tuning.
- **Model merging and compositionality:**
  - Merge multiple LoRA adapters for multi-domain or multi-task capabilities.
  - Compose adapters for new tasks without retraining the base model.

Limitations of Fine-Tuning and RLHF

- **Data Quality:** RLHF relies heavily on high-quality, diverse preference data or human feedback; poor data leads to suboptimal or biased models.
- **Scalability:** Collecting human feedback at scale is expensive and time-consuming.
- **Alignment Gaps:** RLHF may not perfectly capture complex, nuanced human values or intent.
- **Overoptimization:** Models may "game" the reward model, leading to reward hacking or unnatural outputs.
- **Generalization:** Fine-tuning and RLHF can cause models to overfit to specific tasks or feedback distributions, reducing robustness.
- **Catastrophic Forgetting:** Risk of losing previously learned capabilities during aggressive fine-tuning.
- **Computational Cost:** Both fine-tuning and RLHF are resource-intensive, especially for large models.

Limitations of Fine-Tuning and RLHF (cont.)

- **Ethical Concerns:** Human preferences used for training may encode societal biases, under-representation, or other ethical issues.
- **Transparency:** RLHF pipelines can be hard to interpret and debug.

Future Directions

- **RLAIF (RL with AI Feedback):** Automate human labelers using strong AI models for preference data.
- **Constitutional AI:** Use rule-based alignment and self-critique to guide model behavior (e.g., Anthropic's approach).
- **Open RLHF Datasets:** Promote better community sharing and reproducibility with open preference datasets.
- **Multi-modal Alignment:** Extend alignment techniques to vision, audio, and code generation tasks.
- **Reward Hacking Defense:** Develop robust reward functions to prevent models from exploiting reward loopholes.

Summary

- Fine-tuning enables specialization and alignment of LLMs.
- Supervised Fine-Tuning (SFT) is the foundational step.
- RLHF incorporates human preference alignment using PPO.
- DPO streamlines the process with a direct preference loss.
- LoRA and QLoRA make adaptation efficient and accessible.
- The field is rapidly evolving with ongoing innovation.

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